# ERD of the Control Group Consortium

## Institution and People

**institution**
- **institutionId PK**
- institutionName
- institutionType

**department**
- **departmentId PK**
- *institutionId FK*
- departmentName

**payer**
- **payerId PK**
- *personId FK*
- *institutionId FK*
- *departmentId FK*

**affiliation**
- **affiliationId PK**
- *personId FK*
- *institutionId FK*
- validFrom
- validTo

**person**
- **personId PK**
- personName
- email
- phone

## Resources

**resource_type**
- **resourceTypeId PK**
- resourceTypeName
- resourceTypeKind

**qualification**
- **qualificationId PK**
- qualificationName
- qualificationKind

**person_qualification**
- **personQualificationId PK**
- *personId FK*
- *qualificationId FK*
- validFrom
- validTo
- Issuer

**site**
- **siteId PK**
- *institutionId FK*
- siteName
- siteTimeZone

**resource**
- **resourceId PK**
- *resourceTypeId FK*
- *siteId FK*
- resourceName
- serialNumber
- movable
- capability
- limitation
- maxSessionMinutes

**resource_requirement**
- ***resourceId PK FK***
- ***qualificationId PK FK***

**equipment_placement**
- **equipmentPlacementId PK**
- *equipmentId FK*
- *facilityId FK*
- startAt
- endAt

**configuration**
- **configurationId PK**
- *facilityId FK*
- configurationName
- configurationPurpose

## Access Control

**credential**
- **credentialId PK**
- *personId FK*
- *issuerInstitutionId FK*
- externalSubject
- validFrom
- validTo

**role**
- **roleId PK**
- roleName
- canViewConfidential

**access_grant**
- **accessGrantId PK**
- *personId FK*
- *resourceId FK*
- *projectId FK*
- *authorizedBy FK*
- validFrom
- validTo

**person_role**
- ***personId PK FK***
- ***roleId PK FK***

**access_log**
- **accessLogId PK**
- *credentialId FK*
- *resourceId FK*
- *eventId FK*
- occurredAt
- result
- reason

## Project & Request

**project**
- **projectId PK**
- *principalInvestigatorId FK*
- projectCode
- projectTitle
- projectVisibility

**project_member**
- ***projectId PK FK***
- ***personId PK FK***
- responsibility

**experiment**
- **experimentId PK**
- *projectId FK*
- *leadResearcherId FK*
- experimentName
- startAt
- endAt

**request**
- **requestId PK**
- *experimentId FK*
- *requestingInstitutionId FK*
- *requesterId FK*
- submittedAt
- desiredStart
- desiredEnd
- staffingMinutes
- priority
- safetyNotes
- specialConditions
- expectedOutcome

**request_need**
- ***requestId PK FK***
- ***resourceTypeId PK FK***
- quantity
- estimatedMinutes

**decision**
- **decisionId PK**
- *requestId FK*
- *eventId FK*
- *decidedBy FK*
- decidedAt
- outcome
- rationale

## Event

**event**
- **eventId PK**
- *experimentId FK*
- eventType
- startAt
- endAt
- eventStatus

**event_resource**
- ***eventId PK FK***
- ***resourceId PK FK***

**event_configuration**
- ***eventId PK FK***
- ***configurationId PK FK***

**staff_assignment**
- ***eventId PK FK***
- ***personId PK FK***
- duty

**event_change**
- **eventChangeId PK**
- *eventId FK*
- *changedBy FK*
- changedAt
- oldStart
- oldEnd
- oldStatus
- newStart
- newEnd
- newStatus
- reason

**staff_availability**
- **staffAvailabilityId PK**
- *personId FK*
- startAt
- endAt

## Pricing and Allocation

**rate**
- **rateId PK**
- *resourceTypeId FK*
- *consumableId FK*
- staffService
- unitPrice
- billingUnit
- validFrom
- validTo

**consumable**
- **consumableId PK**
- consumableName
- unit

**estimate**
- **estimateId PK**
- *eventId FK*
- *rateId FK*
- estimateQuantity
- estimatedAt

**usage_record**
- **usageId PK**
- *eventId FK*
- *resourceId FK*
- *staffPersonId FK*
- *userId FK*
- *consumableId FK*
- quantity
- startAt
- endAt

**funding_eligibility**
- ***fundingResourceId PK FK***
- ***rateId PK FK***

**funding_resource**
- **fundingResourceId PK**
- *payerId FK*
- fundingResourceName
- fundingResourceType
- budget
- validFrom
- validTo

**policy_share**
- **policyShareId PK**
- *fundingResourceId FK*
- *allocationPolicyId FK*
- *beneficiaryProjectId FK*
- basisPoint

**allocation_policy**
- **allocationPolicyId PK**
- *projectId FK*
- *authorizedBy FK*
- validFrom
- validTo
- rationale

## Billing

**charge**
- **chargeId PK**
- *usageId FK*
- *rateId FK*
- *allocationPolicyId FK*
- state
- createdAt

**adjustment**
- **adjustmentId PK**
- *chargeId FK*
- *authorizedBy FK*
- authorizedAt
- amount
- reason

**charge_allocation**
- **chargeAllocationId PK**
- *chargeId FK*
- *policyShareId FK*
- amount

**invoice_lines**
- ***chargeAllocationId PK FK***
- *invoiceId FK*

**invoice**
- **invoiceId PK**
- *fundingResourceId FK*
- periodStart
- periodEnd
- issuedAt
- status